

# Sri Lanka Institute of Information Technology



## **Web-Based Library Management System**

Group ID: MLB-B11G2-10

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# **1. Introduction**

The Web-Based Library Management System is an online platform developed to manage library operations efficiently. The system allows librarians to manage books, members, borrowing activities, returns, fines, and reports through a centralized web application. Library members can also search books online, check availability, and view their borrowing records.

The system aims to reduce manual paperwork, improve record management, and provide better services for both library staff and members.

## **2. Objectives**

- Manage books and members efficiently.
- Support online book searching and availability checking.
- Record book borrowing and return activities accurately.
- Calculate fines for overdue books automatically.
- Generate reports for library management.

### **Target Users**

#### **Primary Users**

- Library Members
- Librarians

#### **Secondary Users**

- Administrator
- Finance Officer

### **3. Scope**

The system will:

- Register and manage library members.
- Add, update, search, and delete book records.
- Issue books to members.
- Record returned books.
- Calculate fines for late returns.
- Generate operational and financial reports.

#### **Limitations**

- Internet access is required to use the system.
- The system depends on accurate data entered by library staff.
- Physical book damage management is not included in the system.

### **4. Functional Requirements**

#### **1. Member Registration and Login**

- Allow users to create accounts, log in, and manage profiles.

#### **2. Book Management**

- Add, update, search, and delete book records.

#### **3. Borrowing Management**

- Issue books to registered members and record borrowing details.

#### **4. Return Management**

- Record returned books and update book availability.

## 5. Fine Management

- Calculate fines for overdue books and record payments.

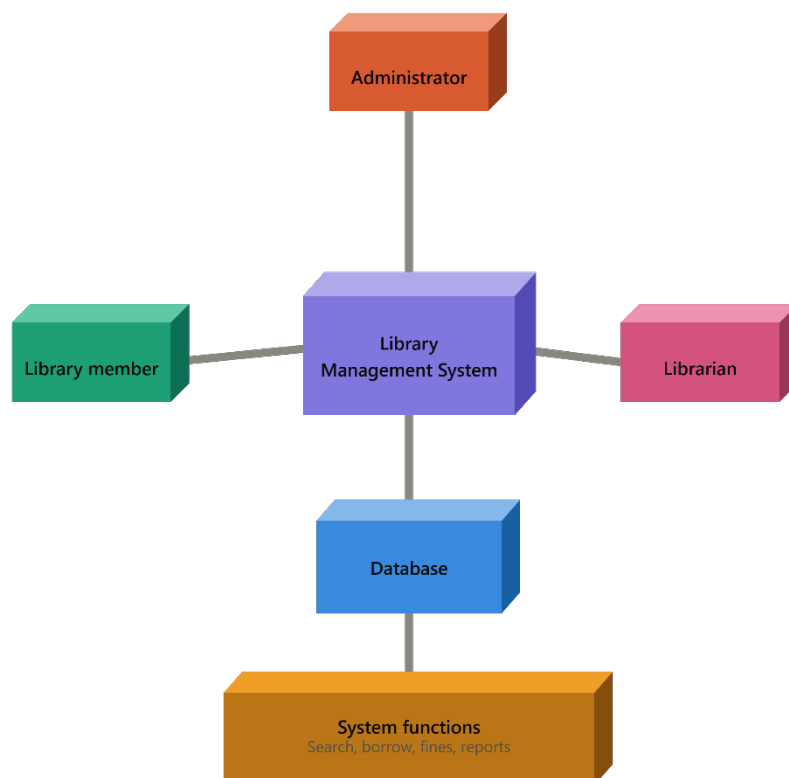
## 6. Online Book Search

- Allow members to search books by title, author, or category.

## 7. Report Generation

- Generate borrowed books, overdue books, and fine collection reports.

## 5. SYSTEM OVERVIEW DIAGRAM



## **6. NON-FUNCTIONAL REQUIREMENTS**

| <b>Requirement</b>     | <b>Description</b>                                                    |
|------------------------|-----------------------------------------------------------------------|
| <b>Performance</b>     | The system should respond quickly and load pages within a few seconds |
| <b>Security</b>        | User passwords should be encrypted and access should be role-based    |
| <b>Usability</b>       | The interface should be simple, clear, and easy to use.               |
| <b>Scalability</b>     | The system should support future growth and additional features       |
| <b>Maintainability</b> | The system should be easy to update and maintain.                     |

## **7. MAJOR STAKEHOLDERS**

| <b>Stakeholder/User</b> | <b>Role</b>                                           |
|-------------------------|-------------------------------------------------------|
| <b>Library Member</b>   | Searches books, borrows books, and checks fines.      |
| <b>Librarian</b>        | Manages books, members, borrowing, and returns.       |
| <b>Administrator</b>    | Manages users, reports, fines, and system settings    |
| <b>Finance Officer</b>  | Views and manages fine payments and financial reports |

## **8. MAJOR FUNCTIONS**

- **User Registration and Login**

| <b>Users</b>                              | <b>Features</b>                                 | <b>Outcome</b>                                  |
|-------------------------------------------|-------------------------------------------------|-------------------------------------------------|
| Members,<br>Librarians,<br>Administrators | Create accounts, log in,<br>and manage profiles | Secure access and<br>personalized<br>experience |

- **Book Management**

| <b>Users</b>                  | <b>Features</b>                          | <b>Outcome</b>                            |
|-------------------------------|------------------------------------------|-------------------------------------------|
| Librarians,<br>Administrators | Add, update, search,<br>and delete books | Accurate and<br>organized book<br>records |

- **Borrowing and Return Management**

| <b>Users</b>        | <b>Features</b>           | <b>Outcome</b>                                                       |
|---------------------|---------------------------|----------------------------------------------------------------------|
| Librarians, Members | Issue and return<br>books | Efficient borrowing<br>process and accurate<br>availability tracking |

- **Fine Management**

| <b>Users</b>        | <b>Features</b>                                                     | <b>Outcome</b>                                          |
|---------------------|---------------------------------------------------------------------|---------------------------------------------------------|
| Librarians, Members | Calculates overdue fines, records payments, and updates fine status | Accurate fine calculation and timely payment management |

- **Online Book Search**

| <b>Users</b>        | <b>Features</b>                                  | <b>Outcome</b>                            |
|---------------------|--------------------------------------------------|-------------------------------------------|
| Members, Librarians | Search books by title, author, ISBN, or category | Quick and easy access to book information |

- **Report management**

| <b>Users</b>               | <b>Features</b>                                                       | <b>Outcome</b>                                               |
|----------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------|
| Administrators, Librarians | Generate reports on borrowed books, overdue books, members, and fines | Better monitoring and decision-making for library operations |

## **9. MINOR FUNCTIONS**

| <b>Minor Function</b>            | <b>Description</b>                             |
|----------------------------------|------------------------------------------------|
| <b>Password Reset</b>            | Allows users to reset forgotten passwords      |
| <b>Profile Management</b>        | Allows users to update personal information    |
| <b>Search and Filter Options</b> | Helps users quickly find books                 |
| <b>View Borrowing History</b>    | Allows members to view previous borrowed books |
| <b>Logout</b>                    | Allows users to securely end their session     |

## **10. SYSTEM LIMITATIONS / CONSTRAINTS**

| <b>Constraint</b>            | <b>Description</b>                       |
|------------------------------|------------------------------------------|
| <b>Web-Based Platform</b>    | Accessible only through a web browser    |
| <b>Internet Dependency</b>   | Requires an active internet connection   |
| <b>Data Accuracy</b>         | Depends on correct data entered by staff |
| <b>Browser Compatibility</b> | Requires modern web browsers             |

### Assumptions

- Users have basic knowledge of web applications.
- Library staff enter accurate book and member information.
- Members use valid contact information for notifications.

## 11. SIMPLE TIMELINE

| Week    | Planned Activities                               |
|---------|--------------------------------------------------|
| Week 3  | Requirement gathering and stakeholder interviews |
| Week 4  | Requirement analysis and finalization            |
| Week 5  | Prepare SRS document                             |
| Week 6  | Design system architecture and database          |
| Week 7  | Develop member management module                 |
| Week 8  | Develop book management module                   |
| Week 9  | Develop borrowing and return module              |
| Week 10 | Develop fine management module                   |
| Week 11 | Develop reporting functions                      |
| Week 12 | Integrate all modules and perform testing        |
| Week 13 | Fix bugs and gather feedback                     |
| Week 14 | Complete documentation and prepare presentation  |

## **12. Conclusion**

The Web-Based Library Management System is designed to improve library operations by automating book management, member management, borrowing, returns, fine calculation, and report generation. The system reduces manual work, improves record accuracy, and provides better services for library members.

Our team plans to complete the project successfully through proper requirement analysis, system design, development, testing, and teamwork. By following the planned timeline, we aim to deliver a secure, reliable, and user-friendly web-based solution.